

Market Risk Research Project

Alan Sieprawski | Contributor: Conor McMillan / 2018-01-02 to 2026-09-18

How did adding bonds and gold change portfolio risk?

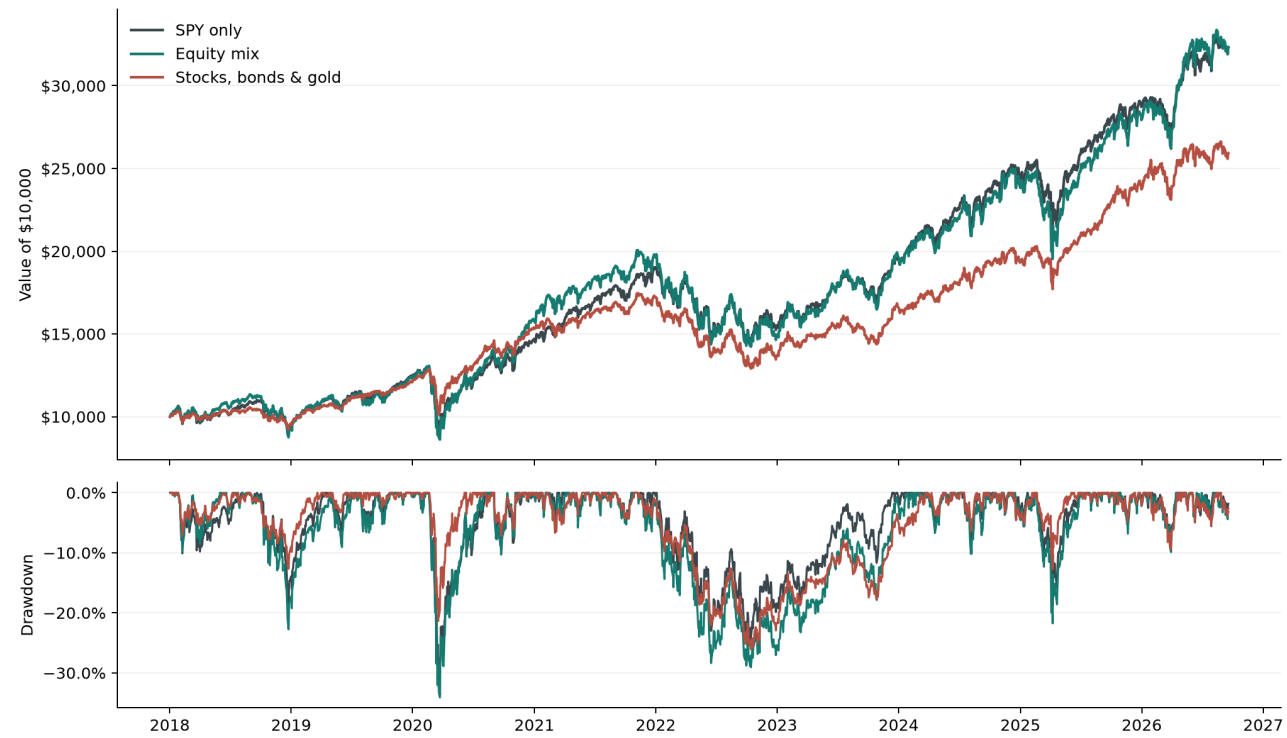
The stocks, bonds and gold mix had 13.6% annualized volatility, 35.6% lower than the equity mix's 21.2%.

Its maximum drawdown was -26.1%, compared with -34.1% for the equity mix and -33.7% for SPY alone.

The diversified mix returned 11.6% annualized, versus 14.4% for the equity mix. The risk and return trade-off both matter.

Portfolio	Ann. return	Volatility	Drawdown	Sharpe
SPY only	14.5%	19.0%	-33.7%	0.81
Equity mix	14.4%	21.2%	-34.1%	0.74
Stocks, bonds & gold	11.6%	13.6%	-26.1%	0.87

What changed when bonds and gold were added?



Monthly rebalancing | 10 bps per dollar bought or sold, including entry | Yahoo adjusted closes | Historical comparison

In 2022 the diversified mix returned -20.8%. Diversification did not guarantee a positive return in that window.

Method and limits

Fixed weights: SPY only (100% SPY); equity mix (one-third SPY, QQQ and IWM); diversified mix (20% each in SPY, QQQ, IWM, TLT and GLD). Weights drift between monthly resets. Costs are 10 bps per dollar bought or sold, including entry; no terminal liquidation. Risk-free rate: assumed 0%. Annualization: 252 trading days. Adjusted closes: Yahoo Finance.

This is a descriptive comparison with chosen assets and periods, not an optimized strategy or a predictive test. Taxes, liquidity and market impact are excluded. Large price moves are retained for review, not independently verified. Results depend on the sample and assumptions.